Prayas JEE AIR (2025)

WEEKLY TEST 21

DURATION : 180 Minutes | **DATE : 19/12/2024** | **M.MARKS : 300**

Topics Covered

Physics:	Full Syllabus
Chemistry:	Full Syllabus
Mathematics:	Full Syllabus

General Instructions:

1. Immediately fill in the particulars on this page of the test booklet.
2. The test is of **3 hours** duration.
3. The test booklet consists of 75 questions. The maximum marks are **300**.
4. There are three Sections in the question paper, Section I, II & III consisting of Section-I **(Physics)**, Section-II **(Chemistry)**, Section-III **(Mathematics)** and having **25 questions** in each part in which first **20** questions are of Objective Type and Last **5 questions** are integers type and all **25 questions are compulsory**.
5. There is only one correct response for each question.
6. Each correct answer will give **4** marks while **1** Mark will be deducted for a wrong response.
7. No student is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, any electronic device, etc. inside the examination room/hall.
8. On completion of the test, the candidate must hand over the Answer Sheet to the Invigilator on duty in the Room/Hall. However, the candidates are allowed to take away this Test Booklet with them.
9. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

OMR Instructions:

1. Use blue/black dark ballpoint pens.
2. Darken the bubbles completely. Don't put a tick mark or a cross mark where it is specified that you fill the bubbles completely. Half-filled or over-filled bubbles will not be read by the software.
3. Never use pencils to mark your answers.
4. Never use whiteners to rectify filling errors as they may disrupt the scanning and evaluation process.
5. Writing on the OMR Sheet is permitted on the specified area only and even small marks other than the specified area may create problems during the evaluation.
6. Multiple markings will be treated as invalid responses.
7. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

Name of the Student (In CAPITALS) : _____

Roll Number: _____

OMR Bar Code Number: _____

Candidate's Signature: _____ Invigilator's Signature _____

SECTION-I (PHYSICS)

Single Correct Type Questions

1. At time t = 0 a particle starts travelling from a height $7\hat{z}$ cm in a plane keeping z-coordinate constant. At any instant of time it's position along the $\hat{x}$ and $\hat{y}$ directions are defined as $3t$ cm and $5t^3$ cm respectively. At t = 1s acceleration of the particle will be

(1) $-30\hat{y}$ cm/sec^2 (2) $30\hat{y}$ cm/sec^2

(3) $3\hat{x}+15\hat{y}$ cm/sec^2 (4) $3\hat{x}+15\hat{y}+7\hat{z}$ cm/sec^2

2. A particle of mass m is moving in the xy-plane such that its velocity at a point (x, y) is given as $\vec{v}=\alpha(y\hat{x}+2x\hat{y})$, where α is a non-zero constant. What is the force $\vec{F}$ acting on the particle?

(1) $\vec{F}=2m\alpha^2(x\hat{x}+y\hat{y})$

(2) $\vec{F}=m\alpha^2(y\hat{x}+2x\hat{y})$

(3) $\vec{F}=2m\alpha^2(y\hat{x}+x\hat{y})$

(4) $\vec{F}=m\alpha^2(x\hat{x}+2y\hat{y})$

3. Given below are two statements:

Statement-I: A truck and a car moving with same kinetic energy are brought to rest by applying brakes which provide equal retarding forces. Both come to rest in equal distance.

Statement-II: A car moving towards east takes a turn and moves towards north, the speed remains unchanged. The acceleration of the car is zero.

In the light of given statements, choose the most appropriate answer from the options given below.

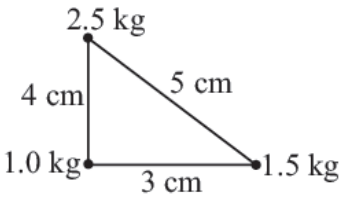
(1) Statement-I is correct but Statement-II is incorrect

(2) Statement-I is incorrect but Statement-II is correct

(3) Both Statement-I and Statement-II are incorrect

(4) Both Statement-I and Statement-II are correct

4. Three point particles of masses 1.0 kg, 1.5 kg and 2.5 kg are placed at three corners of a right angle triangle of sides 4.0 cm, 3.0 cm and 5.0 cm as shown in the figure. The center of mass of the system is at a point:



(1) 0.6 cm right and 2.0 cm above 1kg mass

(2) 2.0 cm right and 0.9 cm above 1 kg mass

(3) 0.9 cm right and 2.0 cm above 1kg mass

(4) 1.5 cm right and 1.2 cm above 1kg mass

5. Match Column–I with Column–II

Column-I		Column-II	
A.	Moment of inertia of solid sphere of mass M and radius R about any tangent.	P	$\frac{5}{3}MR^2$
B.	Moment of inertia of hollow sphere of mass M and radius (R) about any tangent	Q.	$\frac{7}{5}MR^2$
C.	Moment of inertia of circular ring of mass M and radius (R) about its diameter.	R.	$\frac{1}{4}MR^2$
D.	Moment of inertia of circular disc of mass M and radius (R) about any diameter.	S.	$\frac{1}{2}MR^2$

Choose the correct answer from the options given below:

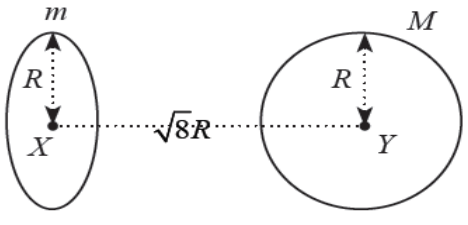
(1) A→Q; B→P; C→S; D→R

(2) A→P; B→Q; C→S; D→R

(3) A→Q; B→P; C→R; D→S

(4) A→P; B→Q; C→R; D→S

6. Find the gravitational force of attraction between the ring and sphere as shown in the diagram, where the plane of the ring is perpendicular to the line joining the centres. If $\sqrt{8}R$ is the distance between the centres of a ring (of mass 'm') and a sphere (mass 'M') where both have equal radius 'R'.



(1) $\frac{\sqrt{8}}{9}\times\frac{GmM}{R}$

(2) $\frac{1}{3\sqrt{8}}\times\frac{GMm}{R^2}$

(3) $\frac{\sqrt{8}}{27}\times\frac{GmM}{R^2}$

(4) $\frac{2\sqrt{2}}{3}\times\frac{GMm}{R^2}$

7. Under the same load, wire *A* having length 5.0 m and cross section 2.5×10^{-5} m^2 stretches uniformly by the same amount as another wire *B* of length 6.0 m and a cross section of 3.0×10^{-5} m^2 stretches. The ratio of the Young's modulus of wire *A* to that of wire *B* will be:

(1) 1 : 4 (2) 1 : 1

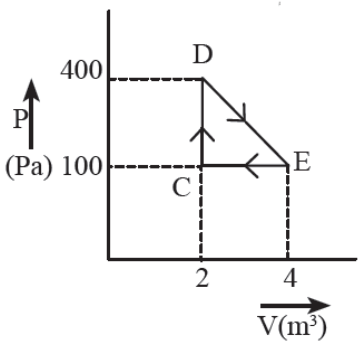
(3) 1 : 10 (4) 1 : 2

8. A small spherical droplet of density *d* is floating exactly half immerse in a liquid ρ and surface tension *T*. The radius of the droplet is (take note that the surface tension applies an upward force on the droplet):

(1) $r = \sqrt{\frac{T}{(d+\rho)g}}$ (2) $r = \sqrt{\frac{3T}{(2d-\rho)g}}$

(3) $r = \sqrt{\frac{T}{(d-\rho)g}}$ (4) $r = \sqrt{\frac{2T}{3(d-\rho)g}}$

9. A thermodynamic system is taken through cyclic process. The total work done in the process is:



(1) 100 J

(2) 300 J

(3) Zero

(4) 200 J

10. In a linear simple harmonic motion (SHM)

(A) Restoring force is directly proportional to the displacement.

(B) The acceleration and displacement are opposite in direction.

(C) The velocity is maximum at mean position.

(D) The acceleration is minimum at extreme points.

Choose the correct answer from the options given below :

(1) (A), (B) and (C) only

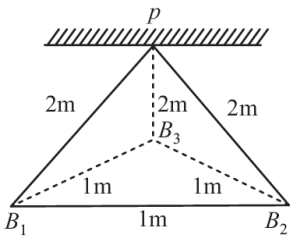
(2) (C) and (D) only

(3) (A), (B) and (D) only

(4) (A), (C) and (D) only

11. Three identical charged balls each of charge 2C are suspended from a common point *P* by silk threads of 2 m each (as shown in figure). They form an equilateral triangle of side 1 m.

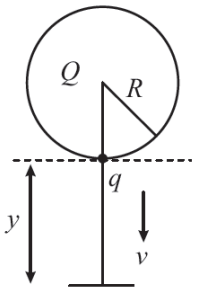
The ratio of net force on a charged ball to the force between any two charged balls will be



(1) 1 : 1 (2) 1 : 4

(3) $\sqrt{3}:2$ (4) $\sqrt{3}:1$

12. A solid sphere of radius *R* carries a charge *Q* + *q* distributed uniformly over its volume. A very small point like piece of it of mass *m* gets detached from the bottom of the sphere and falls down vertically under gravity. This piece carries charge *q*. If it acquires a speed *v* when it has fallen through a vertical height *y* (see figure), then (Assume the remaining portion to be spherical).



(1) $v^2 = y\left[\frac{qQ}{4\pi\varepsilon_0 R(R+y)m} + g\right]$

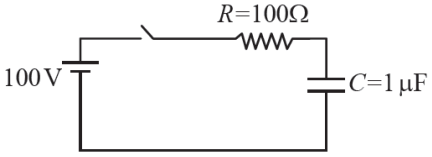
(2) $v^2 = 2y\left[\frac{qQR}{4\pi\varepsilon_0 (R+y)^3 m} + g\right]$

(3) $v^2 = y\left[\frac{qQ}{4\pi\varepsilon_0 R^2 ym} + g\right]$

(4) $v^2 = 2y\left[\frac{qQ}{4\pi\varepsilon_0 R(R+y)m} + g\right]$

13. A capacitor of capacitance $C = 1\mu F$ is suddenly connected to a battery of 100 volt through a resistance $R = 100\ \Omega$. The time taken for the capacitor to be charged to get 50 V is:

[Take $\ell n\,(2) = 0.69$]



(1) 3.33×10^{-4} s

(2) 1.44×10^{-4} s

(3) 0.30×10^{-4} s

(4) 0.69×10^{-4} s

14. As shown in the figure, a long straight conductor with semicircular arc of radius $\frac{\pi}{10}$m is carrying current I = 3A. The magnitude of the magnetic field at the center O of the arc is:
(The permeability of the vacuum $\mu_0 = 4\pi\times10^{-7}$ NA^{-2})

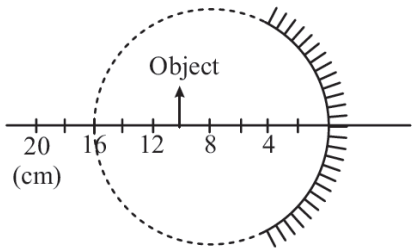


(1) 6μT
(2) 1μT
(3) 4μT
(4) 3μT

15. A square loop of area 25 cm^2 has a resistance of 10Ω. The square loop is placed in a uniform magnetic field of magnitude 40.0 T. The plane of the loop is perpendicular to the magnetic field. The work done in pulling the loop out of the magnetic field slowly and uniformly in 1.0 sec, will be

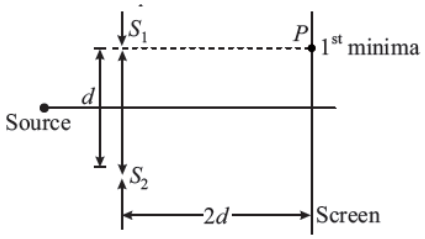
(1) 2.5×10^{-3} J
(2) 1.0×10^{-3} J
(3) 1.0×10^{-4} J
(4) 5×10^{-3} J

16. A spherical mirror is obtained as shown in the figure from a hollow glass sphere. if an object is positioned infront of the mirror, what will be the nature and magnification of the image of the object? (Figure drawn as schematic and not to scale)



(1) Inverted, real and magnified
(2) Erect, virtual and unmagnified
(3) Inverted, real and unmagnified
(4) Erect, virtual and magnified

17. Consider a Young's double slit experiment as shown in figure. What should be the slit separation d in terms of wavelength λ such that the first minima occurs directly in front of the slit (S_1)?



(1) $\frac{\lambda}{2\left(\sqrt{5}-2\right)}$
(2) $\frac{\lambda}{\left(\sqrt{5}-2\right)}$
(3) $\frac{\lambda}{2\left(5-\sqrt{2}\right)}$
(4) $\frac{\lambda}{\left(5-\sqrt{2}\right)}$

18. Identify the correct statements from the following descriptions of various properties of electromagnetic waves.
(A) In a plane electromagnetic wave electric field and magnetic field must be perpendicular to each other and direction of propagation of wave should be along electric field or magnetic field.
(B) The energy in electromagnetic wave is divided equally between electric and magnetic fields.
(C) Both electric field and magnetic filed are parallel to each other and perpendicular to the direction of propagation of wave.
(D) The electric field, magnetic field and direction of propagation of wave must be perpendicular to each other.
(E) The ratio of amplitude of magnetic field to the amplitude of electric field is equal to speed of light.
Choose the most appropriate answer from the options given below:
(1) (D) only
(2) (B) and (D) only
(3) (B), (D) and (E) only
(4) (A), (B) and (E) only

19. The angular momentum for the electron in Bohr's orbit is integer multiple of L. If the electron is assumed to revolve in second orbit of hydrogen atom for a very short duration, then the change in angular momentum will be:

(1) $\frac{L}{2}$ (2) Zero
(3) L (4) $2L$

20. In the following nuclear reaction,

$$D \xrightarrow{\alpha} D_1 \xrightarrow{\beta^-} D_2 \xrightarrow{\alpha} D_3 \xrightarrow{\gamma} D_4$$

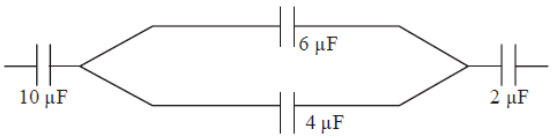
Mass number of D is 182 and atomic number is 74. Mass number and atomic number of D_4 respectively will be ____

(1) 174 and 71 (2) 174 and 69
(3) 172 and 69 (4) 172 and 71

Integer Type Questions

21. Equation of a stationary and a travelling waves are as follows $y_1 = a\sin(kx)\cos(\omega t)$ and $y_2 = a\sin(\omega t - kx)$. The phase difference between two points $x_1 = \frac{\pi}{3k}$ and $x_2 = \frac{3\pi}{2k}$ is ϕ_1 in the standing wave (y_1) and is ϕ_2 in travelling wave (y_2) then find the value of $\frac{7\phi_1}{\phi_2}$.

22. In the figure shown below, the charge on the left plate of the 10 μF capacitor is –30μC. The charge (in μC) on the right plate of the 6μF capacitor is:



23. A series *AC* circuit containing an inductor (20mH), a capacitor (100μF) and a resistor (20Ω) is driven by an *AC* source of 25V and $\frac{500}{\pi}$ Hz. The energy (in Joule) dissipated in the circuit in 6 sec is:

24. Surface of certain metal is first illuminated with light of wavelength $\lambda_1 = 124\text{nm}$ and then, by light of wavelength $\lambda_2 = 248\text{nm}$. It is found that the maximum speed of the photo electrons in the two cases differ by a factor of (2). The work function of the metal (in eV) is ϕ. Than find the value of 3ϕ

(Energy of photon = $\frac{1240}{\lambda\ (\text{in nm})}$ eV)

25. The coefficient of apparent volume expansion of mercury in a glass vessel is 153×10^{-6}/°C and in a steel vessel is 144×10^{-6}/°C. If linear coefficient of thermal expansion for steel is 12×10^{-6}/°C, then linear coefficient of thermal expansion of glass is $x \times 10^{-6}$ /°C. Find the value of x.

SECTION-II (CHEMISTRY)

Single correct Type Questions

26. Which of the following statement is wrong?

(1) Nitrogen cannot form dπ - pπ bond.
(2) Single N - N bond is weaker than the single P – P bond,
(3) N_2O_4 contain N–N bond.
(4) The stability of hydrides increases from NH_3 to BiH_3 in group 15 of the periodic table

27. The molar solubility of $Cd(OH)_2$ is 1.84×10^{-5} M in water. The expected solubility of $Cd(OH)_2$ in a buffer solution of pH = 12 is:

(1) 6.23×10^{-11} M (2) 1.84×10^{-9} M
(3) $\frac{2.49}{1.84} \times 10^{-9}$ M (4) 2.49×10^{-10} M

28. Which of the glycosidic linkage between galactose and glucose is present in lactose?

(1) C–1 of glucose and C–6 of galactose
(2) C–1 of galactose and C–4 of glucose
(3) C–1 of glucose and C–4 of galactose
(4) C–1 of galactose and C–6 of glucose

29. Calculate the standard cell potential (in V) of the cell in which following reaction takes place:

$Fe^{2+}(aq) + Ag^{+}(aq) \rightarrow Fe^{3+}(aq) + Ag(s)$

Given that

$E^{\circ}_{Ag^+/Ag} = xV;\ E^{\circ}_{Fe^{2+}/Fe} = yV;\ E^{\circ}_{Fe^{3+}/Fe} = zV$

(1) x – y
(2) x + y – z
(3) x + 2y – 3z
(4) x – z

30. Ethylamine ($C_2H_5NH_2$) can be obtained from N-ethyl phthalimide on treatment with:

(1) $NaBH_4$
(2) CaH_2
(3) H_2O
(4) NH_2NH_2

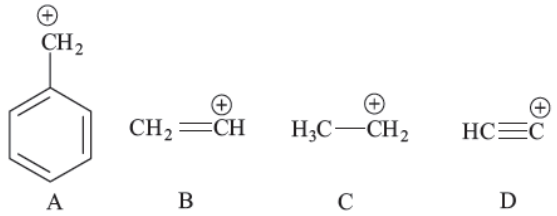
31. The correct order of ionic radii for the ions, P^{3-}, S^{2-}, Ca^{2+}, K^+, Cl^- is:

(1) $K^+ > Ca^{2+} > P^{3-} > S^{2-} > Cl^-$
(2) $P^{3-} > S^{2-} > Cl^- > Ca^{2+} > K^+$
(3) $Cl^- > S^{2-} > P^{3-} > Ca^{2+} > K^+$
(4) $P^{3-} > S^{2-} > Cl^- > K^+ > Ca^{2+}$

32. A reaction of benzonitrile with one equivalent CH_3MgBr followed by hydrolysis produces a yellow liquid (P). The compound (P) will give positive:

(1) Iodoform test (2) Schiff's test
(3) Ninhydrin's test (4) Tollen's test

33.

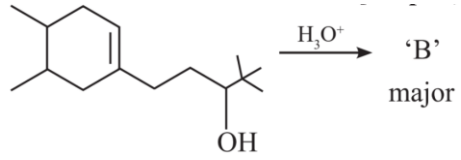


A B C D

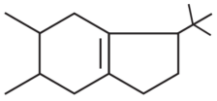
The correct order of stability of given carbocations is:

(1) A > C > B > D (2) D > B > C > A
(3) C > A > D > B (4) D > B > A > C

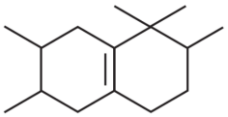
34. In the following reaction, 'B' is



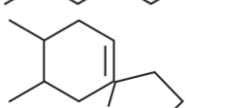
(1)



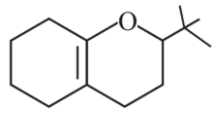
(2)



(3)



(4)



35. Match **Column-** I with **Column**– II

Column- I (Name of oxo acid)		**Column-II (Oxidation state of "P")**	
A	Hypophosphorous acid	P	+ 5
B	Orthophosphoric acid	Q	+ 4
C	Hypophosphoric acid	R	+ 3
D	Orthophosphorous acid	S	+ 2
		T	+ 1

Choose the correct answer from the options given below:

(A) A→T; B→S; C→Q; D→R
(B) A→S; B→P; C→Q; D→R
(C) A→S; B→T; C→Q; D→R
(D) A→T; B→P; C→Q; D→R

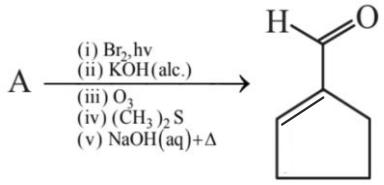
36. 250 g solution of D-glucose in water contains 10.8% of carbon by weight. The molality of the solution is nearest to (Given : Atomic weights are, H – 1u ; C – 12u ; O – 16u)

(1) 1.03 m
(2) 2.06 m
(3) 3.09 m
(4) 5.40 m

37. The time for half life period of a certain reaction A → products is 1 hour. When the initial concentration of the reactant 'A', is 2.0 mol L^{-1}, how much time does it take for its concentration to come from 0.50 to 0.25 mol L^{-1} if it is a zero order reaction?

(1) 4h
(2) 0.5 h
(3) 0.25 h
(4) 1 h

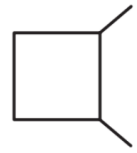
38. In the following reaction A is



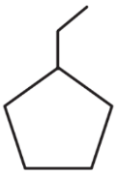
(1)



(2)



(3)



(4)



39. The set having ions which are coloured and paramagnetic both is:

(1) Ni^{2+}, Mn^{7+}, Hg^{2+}
(2) Cu^{+}, Zn^{2+}, Mn^{4+}
(3) Sc^{3+}, V^{5+}, Ti^{4+}
(4) Cu^{2+}, Cr^{3+}, Ni^{2+}

40. In order to oxidise a mixture containing one mole each of FeC_2O_4, $Fe_2(C_2O_4)_3$, $FeSO_4$ and $Fe_2(SO_4)_3$ in acidic medium, the number of moles of $KMnO_4$ required is:

(1) 3
(2) 2
(3) 1
(4) 1.5

41. Two complexes(I) $[Cr(H_2O)_6]Cl_3$ and (II) $[Cr(NH_3)_6]Cl_3$ are violet and yellow coloured respectively. The incorrect statement regarding them is

(1) Δ_0 values of (I) and (II) are calculated from the energies of violet and yellow light, respectively.

(2) both are paramagnetic with three unpaired electrons

(3) both absorbs energies corresponding to their complementary colours

(4) Δ_0 value for (I) is less than that of (II)

42. Given below are two statements:

Statement I: C_2H_5OH and AgCN both can generate nucleophile.

Statement II: KCN and AgCN both will generate nitrile nucleophile with all reaction conditions.

Choose the most appropriate option:

(1) Both statement I and statement II are false.

(2) Statement I is false but statement II is true.

(3) Both statement I and statement II are true.

(4) Statement I is true but statement II is false.

43. The number of subshells associated with n = 4 and m = –2 quantum numbers is:

(1) 2 (2) 16

(3) 8 (4) 4

44. A solution containing 2.675 g of $CoCl_3.6NH_3$ (molar mass = 267.5 g mol^{-1}) is passed through a cation exchanger. The chloride ions obtained in solution were treated with excess of $AgNO_3$ to give 4.78 g of AgCl (molar mass = 143.5 g mol^{-1}). The formula of the complex is (At. Mass of Ag = 108 u)

(1) $[Co(NH_3)_6]Cl_3$ (2) $[CoCl_2(NH_3)_4]Cl$

(C) $[CoCl_3(NH_3)_3]$ (D) $[CoCl(NH_3)_5]Cl_2$

45. Ozonolysis of an organic compound gives formaldehyde as one of the products. This confirms the presence of:

(1) a vinyl group

(2) an isopropyl group

(3) an acetylenic triple bond

(4) two ethylenic double bonds

Integer type Questions

46. $2NO + 2H_2 \rightarrow N_2 + 2H_2O$

The above reaction has been studied at 800ºC. The related data are given in the table below.

Reaction serial number	**Initial Pressure of H_2 / kPa**	**Initial Pressure of NO / kPa**	**Initial rate $\left(\frac{-dp}{dt}\right)$ / (kPa / s)**
1	65.5	40.0	0.135
2	65.6	20.1	0.033
3	38.6	65.6	0.214
4	19.2	65.6	0.106

The order of the reaction with respect to NO is_____.

47. The number of bent-shaped molecule/s from the following is _____.

N_3^-, NO_2^-, I_3^-, O_3, SO_2

48. 80 mole percent of $MgCl_2$ is dissociated in aqueous solution. The vapour pressure of 1.0 molal aqueous solution of $MgCl_2$ at 38°C is _____ mm Hg. (Nearest Integer)

Given : Vapour pressure of water at 38°C is 50 mmHg

49. The number of stereoisomers possible for a compound of the molecular formula $CH_3 – CH = CH – CH(OH) – CH_3$ is:

50. Five moles of an ideal gas at 293 K is expanded isothermally from an initial pressure of 2.1 MPa to 1.3 MPa against at constant external pressure 4.3 MPa. The heat transferred in this process is_____kJ mol^{-1}. (Rounded-off to the nearest integer). [Use R= 8.314 J mol^{-1} K^{-1}]

SECTION-III (MATHEMATICS)

Single Correct Type Questions

51. Let $\lambda \neq 0$ be in R. If α and β are the roots of the equation, $x^2 - x + 2\lambda = 0$ and α and γ are the roots of the equation, $3x^2 - 10x + 27\lambda = 0$, then $\frac{\beta\gamma}{\lambda}$ is equal to:

(1) 18 (2) 36

(3) 9 (4) 27

52. Let α be a root of the equation $(a - c)x^2 + (b - a)x + (c - b) = 0$ where a, b, c are distinct real numbers such that the matrix $\begin{bmatrix} \alpha^2 & \alpha & 1 \\ 1 & 1 & 1 \\ a & b & c \end{bmatrix}$ is singular. Then the value of

$$\frac{(a-c)^2}{(b-a)(c-b)} + \frac{(b-a)^2}{(a-c)(c-b)} + \frac{(c-b)^2}{(a-c)(b-a)} \text{ is}$$

(1) 6 (2) 3
(3) 9 (4) 12

53. Let $\left(2x^2 + 3x + 4\right)^{10} = \sum_{r=0}^{20} a_r x^r$. Then $\frac{a_7}{a_{13}}$ is equal to

(1) 2 (2) 4
(3) 6 (4) 8

54. For a complex number z, let Re (z) denote the real part of z. Let S be the set of all complex numbers z satisfying $z^4 - |z|^4 = 4iz^2$, where $i = \sqrt{-1}$. Then the minimum possible value of $|z_1 - z_2|^2$, where $z_1, z_2 \in S$ with Re $(z_1) > 0$ and Re $(z_2) < 0$, is

(1) 3 (2) 8
(3) 12 (4) 16

55. Let R be a rectangle given by the lines $x = 0$, $x = 2$, $y = 0$ and $y = 5$. Let $A(\alpha, 0)$ and $B(0, \beta)$, $\alpha \in [0, 2]$ and $\beta \in [0, 5]$, be such that the line segment AB divides the area of the rectangle R in the ratio 4 : 1. Then, the mid-point of AB lies on a

(1) parabola (2) hyperbola
(3) straight line (4) circle

56. Consider the lines L_1 and L_2 given by

$$L_1 : \frac{x-1}{2} = \frac{y-3}{1} = \frac{z-2}{2}$$

$$L_2 : \frac{x-2}{1} = \frac{y-2}{2} = \frac{z-3}{3}$$

A line L_3 having direction ratios 1, –1, –2, intersects L_1 and L_2 at the points P and Q respectively. Then the length of line segment PQ is

(1) $2\sqrt{6}$ (2) $3\sqrt{2}$
(3) $4\sqrt{3}$ (4) 4

57. Consider the following frequency distribution:

Class	10-20	20-30	30-40	40-50	50-60
Frequency	α	110	54	30	β

If the sum of all frequencies is 584 and median is 45, then $|\alpha - \beta|$ is equal to.

(1) 123 (2) 164
(3) 154 (4) 132

58. If $a_1, a_2, a_3, \ldots a_n$ are in A.P. and $a_1 + a_4 + a_7 + \cdots + a_{16} = 114$, then $a_1 + a_6 + a_{11} + a_{16}$ is equal to:

(1) 38 (2) 98
(3) 76 (4) 64

59. Let $A(0, 1)$, $B(1, 1)$ and $C(1, 0)$ be the mid - points of the sides of a triangle with incentre at the point D. If the focus of the parabola $y^2 = 4ax$ passing through D is $(\alpha + \beta\sqrt{2}, 0)$, where α and β are rational numbers, then $\frac{\alpha}{\beta^2}$ is equal to

(1) 6
(2) 8
(3) 12
(4) $\frac{9}{2}$

60. Let $f, g : R \to R$ be two real valued functions defined as

$$f(x) = \begin{cases} -|x+3| & , x < 0 \\ e^x & , x \geq 0 \end{cases} \text{ and}$$

$$g(x) = \begin{cases} x^2 + k_1 x & , x < 0 \\ 4x + k_2 & , x \geq 0 \end{cases}$$

where k_1 and k_2 are real constants. If (gof) is differentiable at $x = 0$, then $(gof)(-4) + (gof)(4)$ is equal to

(1) $4(e^4 + 1)$
(2) $2(2e^4 + 1)$
(3) $4e^4$
(4) $2(2e^4 - 1)$

61. If the number of words, with or without meaning, which can be made using all the letters of the word MATHEMATICS in which C and S do not come together, is (6!)k, then k is equal to

(1) 1890 (2) 945
(3) 2835 (4) 5670

62. If $\cot^{-1}(\alpha) = \cot^{-1}2 + \cot^{-1}8 + \cot^{-1}18 + \cot^{-1}32 + \cdots$ upto 100 terms, then α is:

(1) 1.02 (2) 1.33
(3) 1.01 (4) 1.00

63. The minimum number of elements that must be added to the relation $R = \{(a, b), (b, c), (b, d)\}$ on the set $\{a, b, c, d\}$ so that it is an equivalence relation, is

(1) 10
(2) 12
(3) 11
(4) 13

64. If circumcenter of an equilateral triangle inscribed in $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ with vertices having eccentric angles α, β, γ respectively is (x_1, y_1), then $\sum\cos\alpha\cos\beta+\sum\sin\alpha\sin\beta$ is

(1) $\frac{9x_1^2}{a^2}+\frac{9y_1^2}{b^2}+\frac{3}{2}$ (2) $9x_1^2-9y_1^2+a^2b^2$

(3) $\frac{9x_1^2}{2a^2}+\frac{9y_1^2}{2b^2}-\frac{3}{2}$ (4) $\frac{9x_1^2}{a^2}+\frac{9y_1^2}{b^2}+3$

65. Let $\vec{a}=5\hat{i}-\hat{j}-3\hat{k}$ and $\vec{b}=\hat{i}+3\hat{j}+5\hat{k}$ be two vectors. Then which one of the following statements is TRUE?

(1) Projection of $\vec{a}$ on $\vec{b}$ is $\frac{17}{\sqrt{35}}$ and the direction of the projection vector is same as of $\vec{b}$

(2) Projection of $\vec{a}$ on $\vec{b}$ is $\frac{-13}{\sqrt{35}}$ and the direction of the projection vector is same as of $\vec{b}$

(3) Projection of $\vec{a}$ on $\vec{b}$ is $\frac{14}{\sqrt{35}}$ and the direction of the projection vector is opposite to the direction of $\vec{b}$

(4) Projection of $\vec{a}$ on $\vec{b}$ is $\frac{-15}{\sqrt{35}}$ and the direction of the projection vector is opposite to the direction of $\vec{b}$

66. The probability that two randomly selected subsets of the set {1, 2, 3, 4, 5} have exactly two elements in their intersection is:

(1) $\frac{65}{2^7}$ (2) $\frac{35}{2^7}$

(3) $\frac{135}{2^9}$ (4) $\frac{65}{2^8}$

67. The differential equation of the family of curves, $x^2 = 4b(y + b)$, $b\in R$, is

(1) $x(y')^2 = x-2yy'$

(2) $xy'' = y'$

(3) $x(y')^2 = x + 2yy'$

(4) $x(y')^2 = 2yy'-x$

68. The area of the region bounded by $y-x = 2$ and $x^2=y$ is equal to:

(1) $\frac{2}{3}$ (2) $\frac{9}{2}$

(3) $\frac{16}{3}$ (4) $\frac{4}{3}$

69. Consider the function $f : R \to R$ defined by $f(x)=\begin{cases}\left(2-\sin\left(\frac{1}{x}\right)\right)|x|, & x\neq 0\\ 0, & x=0\end{cases}$. Then f is:

(1) Monotonic on $(-\infty, 0)$ only

(2) Monotonic on $(-\infty, 0)\cup(0, \infty)$

(3) Monotonic on $(0, \infty)$ only

(4) Not monotonic on $(-\infty, 0)$ and $(0, \infty)$

70. The remainder when $(2021)^{2022} + (2022)^{2021}$ is divided by 7 is

(1) 0

(2) 1

(3) 2

(4) 6

Integer Type Questions

71. Let $f(x)=\max\{|x+1|,|x+2|,\ldots,|x+5|\}$. The $\int_{-6}^{0} f(x)dx$ is equal to

72. Consider a circle $C_1: x^2 + y^2 - 4x - 2y = \alpha - 5$. Let its mirror image in the line $y = 2x + 1$ be another circle $C_2: 5x^2 + 5y^2 - 10fx - 10gy + 36 = 0$. Let r be the radius of C_2. Then $\alpha + r$ is equal to____.

73. If $\int\frac{2e^x+3e^{-x}}{4e^x+7e^{-x}}dx=\frac{1}{14}\left(ux+v\log_e\left(4e^x+7e^{-x}\right)\right)+C$ where C is a constant of integration, then $u + v$ is equal to_____.

74. Let $H_n=\frac{x^2}{1+n}-\frac{y^2}{3+n}=1, n\in N$. Let k be the smallest even value of n such that the eccentricity of H_k is a rational number. If l is length of the latus return of H_k, then $21l$ is equal to

75. $96\cos\frac{\pi}{33}\cos\frac{2\pi}{33}\cos\frac{4\pi}{33}\cos\frac{8\pi}{33}\cos\frac{16\pi}{33}$ is equal to



PW Web/App - https://smart.link/7wwosivoicgd4

Library- https://smart.link/sdfez8ejd80if